VITYARTHI

INTRODUCTION TO

AI & MACHINE LEARNING

(CSA2001)

PROJECT REPORT

Predicting Housing Trends Using Linear Regression

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ACADEMIC YEAR: 2025-26

1. Introduction

In the modern real estate market, **guessing the price** of a home is no longer enough. Data-driven valuation is at the heart of companies like Zillow and Redfin. For this project, I built a **Machine Learning model** to explore how specific economic factors—primarily **median income**—directly impact the market value of houses in California.

The core objective was to move beyond simple observation and use **mathematical modeling** to predict future pricing trends.

2. The Dataset

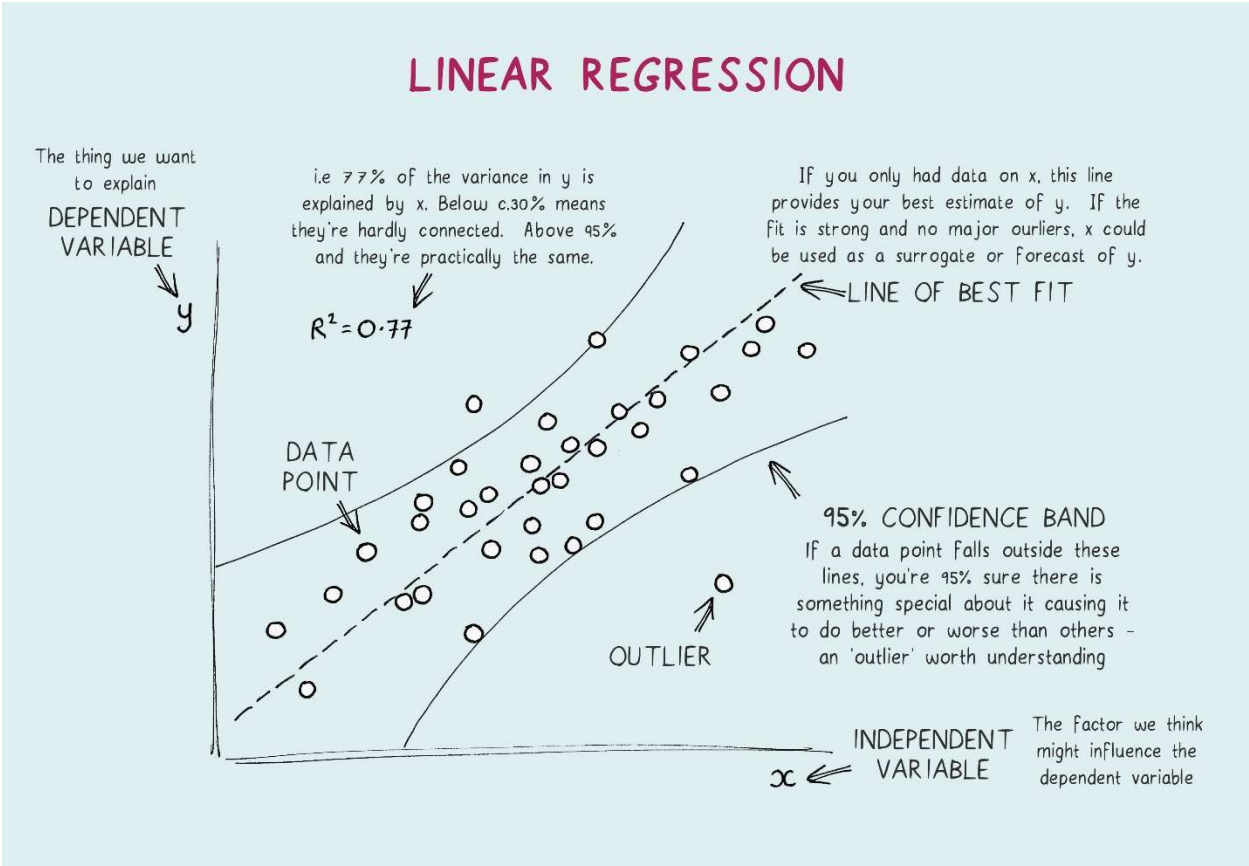
I utilized the **California Housing Dataset**, a standard benchmark in the ML community. Unlike "toy" datasets, this contains real-world data from the 1990 Census, covering over 20,000 California districts.

- **Key Variable:** `MedInc` (Median income in a block group).
- **Target:** `MedHouseVal` (The actual price of the home).

3. Methodology: My Approach

I followed a **standard data science pipeline** to ensure the results were reliable:

- **Data Preparation:** I used the Pandas library to structure the data. I focused on income because, **intuitively**, the wealthier an area is, the higher the property values tend to be.
- **The Split:** A common mistake in ML is testing a model on the same data it learned from. To avoid this, I **segregated the data** into an 80% training set and a 20% testing set.
- **The Algorithm:** I chose **Linear Regression**. It's the "bread and butter" of predictive modeling because it calculates the **best-fit line** that minimizes the distance between predicted and actual prices.



4. Results & Analysis

After training the model, I evaluated it using two main metrics: **R2 Score** and **Mean Squared Error (MSE)**.

- **Observations:** The model produced a clear **positive correlation**. As the income of the residents increased, the price of the houses followed a predictable upward path.
- **The "Human" Element:** While the math is precise, the data shows **outliers**. Some houses are priced much higher than the income suggests—likely due to factors the model doesn't see yet, like **ocean views** or historical significance.

5. Challenges Faced

One hurdle I encountered was setting up the **Python environment**. Initially, I faced *ModuleNotFound* errors in VS Code, which I resolved by managing my **package installations** via pip. Transitioning the project to **Google Colab** eventually allowed for a smoother workflow and better visualization of the graphs.

6. Conclusion

This project served as a **foundational deep dive** into Supervised Learning. It proved that even a "simple" linear model can provide **meaningful insights** into complex markets. Moving forward, I plan to improve this model by adding more "features" like the number of rooms and the age of the house to see if the **prediction accuracy** increases.

7. References

1. Official Library Documentation (The "Primary" Sources)

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